

Chandra Kiran Narala

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SUMMARY

Software Engineer with experience building distributed data pipelines, high-throughput microservices, and performance-critical systems in Python and C++. Strong background in scalable system design, fault-tolerant architectures, and algorithm optimization, with a track record of shipping production systems under real-time constraints

EDUCATION

NYU Tandon | MS in Electrical Engineering | USA

Jan 2022 - May 2024

Courses: Computer System Architecture, Advanced VLSI, Advanced Hardware Design, DSP

GVPCOE | Bachelors in Electronics and Communications Engineering | India

Jul 2016 - Sept 2020

Courses: Data Structures & Algorithms, System Design & Optimization, Computer Architecture, Digital Signal Processing, OS

SKILLS

Programming: C, C++ (Modern C++), Python (PyTorch, NumPy, Pandas, SciPy), Verilog, SystemVerilog

Development Tools: Git, Linux, MySQL, Cocotb/Pytest for verification, Selenium, Xilinx Vivado, Matlab, Cadence Virtuoso/ Innovus

Systems & Methodologies: Object-Oriented Design, Design Patterns, Multithreading, Distributed Systems, Agile / Scrum

Protocols & Techniques: Ethernet (TCP/IP), system-level optimization, Data Processing Pipelines, Fault Tolerant Systems

PROJECTS

Python Real-Time Transit Data Aggregation Service

- Architected a Python-based system that integrates live arrival data from 7 Metropolitan Transportation Authority feeds across 22 subway lines, parsing binary protobuf data and delivering per-direction schedules with sub-1s latency at 400 - 500 daily requests
- Built optimized feed-routing logic mapping each train line to its source endpoint, reducing unnecessary API calls by up to 85% on targeted queries, and deployed on Render with fault-tolerant recovery and zero-downtime restarts

C++-Based Huffman Compression System

- Executed a C++ compression utility using Huffman coding with memory management, bit-level operations, and efficient file I/O
- Developed a high-performance file compression tool implementing Huffman coding algorithm for optimal data storage solutions
- Achieved 47% size reduction with 1.89:1 compression ratio and sub-second processing for 2MB files. Delivered perfect lossless reconstruction with 0.28s encoding and 0.37s decoding performance

Biometric Fingerprint Identification System

- Developed a biometric fingerprint identification system using Python and OpenCV that processes raw fingerprint images through a 3-stage pipeline including background removal, ridge enhancement, and SIFT feature extraction
- Implemented FLANN-based matching algorithm with geometric verification using RANSAC, achieving greater than 90% accuracy with 15+ keypoint correspondences and reducing false positives by 40%
- Built complete end-to-end solution with Tkinter GUI for real-time processing, processing comparisons in ~4 seconds per fingerprint

Distributed Ultra-Low Latency FPGA-Based FIX Parser

- Engineered a performance optimized message parsing engine (C++/ Verilog) on FPGA for ultra-low latency FIX parsing in HFTs
- Designed a custom FSM-driven design featuring a round-robin FIFO buffer for parallel message storage and precompiled BRAM lookup tables for rapid tag=value mapping and real-time validation, thereby eliminating the runtime overhead of XML parsing
- Projected to achieve up to 200 times faster processing; delivering latencies in 50µs range and handling millions of messages per second compared to CPU-based parsing, with 7ms average latency

PROFESSIONAL EXPERIENCE

SRAM DV Engineer | NYU, USA

Jan 2024 - Nov 2025

- Led a 4-person research team to optimize a 256×4-bit SRAM array in 7 nm FinFET, using mathematical modeling and statistical analysis to reduce power consumption by 20% compared to 6T SRAM cell
- Developed and verified SRAM testbenches using Cocotb, enabling Python-based verification with functional and latency testing
- Executed Timing Analysis to achieve timing closure while optimizing performance, area, and power trade-offs. Resolved critical timing violations for reliable read and write operations at higher clock frequencies, optimizing memory access stability
- Collaborated with physical design team on RTL-level fixes during timing closure, hands-on exposure to Cadence ASIC flow

Software Engineer | Cognizant Technology Solutions, India

Nov 2020 - Dec 2021

- Optimized distributed data processing pipeline handling 100K+ requests/second using Python multiprocessing and connection pooling, reducing latency 20% and increasing throughput 15% for 50K+ daily active users
- Built fault-tolerant microservices with circuit breakers and retry mechanisms achieving 99.9% uptime; implemented feature toggles and A/B testing for data-driven rollouts while managing ambiguous requirements
- Automated Python-based deployment pipeline reducing deployment time by 40% through parallel testing and smart test selection; accelerated feature delivery 40% while maintaining 80% code coverage
- Worked cross-functionally with Quality Assurance and software teams to deploy performance patches in agile sprints